

# Project Overview

K Star Fintech — System Architecture & Feature Summary

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## 1. Project Introduction

K Star Fintech is a full-stack microservices-based financial technology application. It provides users with a comprehensive digital banking experience including multi-currency wallets, real-time currency exchange, UPI P2P transfers powered by Kafka event streaming, NFT minting and trading, and ACH/NACHA bank payment processing.

The project is designed with modern enterprise-grade architecture patterns: each business domain is implemented as an independent microservice, all services communicate asynchronously through Apache Kafka, and the frontend is a React single-page application styled with Material UI.

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## 2. Feature Summary

| Feature               | Description                                  | Service          | Status  |
|-----------------------|----------------------------------------------|------------------|---------|
| Registration & Login  | JWT-based auth with BCrypt password hashing  | auth-service     | ■ Ready |
| Multi-Currency Wallet | INR, USD, EUR, GBP, JPY per user accounts    | accounts-service | ■ Ready |
| Real-Time Exchange    | Live rate conversion between 5 currencies    | exchange-service | ■ Ready |
| UPI P2P Transfer      | Event-driven P2P payments via Kafka          | transfer-service | ■ Ready |
| NACHA/ACH Payment     | US ACH bank payments (PPD, CCD, WEB)         | acha-service     | ■ Ready |
| NFT Marketplace       | Mint, list, buy, and trade digital assets    | nft-service      | ■ Ready |
| Dashboard & Charts    | Balance history, portfolio pie, activity bar | frontend         | ■ Ready |
| Kafka Event Streaming | Async events for transfers, NFTs, NACHA      | cloud-services   | ■ Ready |
| Gateway & Auth Filter | Central routing with JWT validation          | gateway-service  | ■ Ready |

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## 3. System Architecture

The application follows a microservices architecture with an API Gateway as the single entry point. The Gateway validates JWT tokens and routes requests to downstream services. Each service is independently deployable via Docker.

| Component        | Port  | Database | Technology                                  |
|------------------|-------|----------|---------------------------------------------|
| API Gateway      | 8080  | None     | Spring Cloud Gateway (WebFlux)              |
| User Service     | 8081  | MySQL    | Spring Boot + JPA + Kafka Producer          |
| Exchange Service | 8082  | MongoDB  | Spring Boot + Mongo + Scheduled Rates       |
| Transfer Service | 8083  | MySQL    | Spring Boot + JPA + Kafka Consumer/Producer |
| NFT Service      | 8084  | MongoDB  | Spring Boot + MongoDB                       |
| NACHA Service    | 8085  | MySQL    | Spring Boot + JPA + Kafka                   |
| Frontend         | 3000  | None     | React 18 + Material UI + Recharts           |
| MySQL            | 3306  | —        | Docker: mysql:8.0                           |
| MongoDB          | 27017 | —        | Docker: mongo:6.0                           |
| Kafka            | 9092  | —        | Confluent cp-kafka:7.4.0                    |

## 4. Technology Stack

### ■ Backend

- Java 17 — LTS version with record types, sealed classes, text blocks
- Spring Boot 3.1.5 — Auto-configuration, embedded server, actuator
- Spring Cloud Gateway — Reactive API gateway with custom filters
- Spring Data JPA + Hibernate — ORM for MySQL relational data
- Spring Data MongoDB — Document storage for exchange rates and NFTs
- Spring Kafka — Producer/Consumer event streaming
- Spring Security — Stateless JWT authentication
- Lombok — Boilerplate reduction (@Data, @Builder, @RequiredArgsConstructor)
- jjwt (io.jsonwebtoken) — JWT token generation and validation
- Maven — Dependency management and build lifecycle

### ■ Frontend

- React 18 — Hooks-based functional components with Concurrent Mode
- React Router v6 — Declarative client-side routing with protected routes
- Material UI v5 — Component library with custom dark theme
- Recharts — Composable chart library (AreaChart, PieChart, BarChart)
- Axios — HTTP client with JWT interceptor and 401 redirect
- Context API — Global auth state management without Redux

### ■ Infrastructure

- Docker + Docker Compose — Containerized infrastructure orchestration
- MySQL 8.0 — Relational database for users, wallets, transactions, NACHA payments
- MongoDB 6.0 — Document database for exchange rates, NFTs, conversion history

- Apache Kafka (Confluent 7.4) — Event streaming backbone
- Zookeeper — Kafka cluster coordination